

Department of Computer Science and Engineering, SVNIT, Surat.
End Semester Examinations, May 2025
B.Tech. III (CSE), Semester - VI
Course: Cloud Computing (CS304)

Date: 06/05/2025

Time: 09:00hrs to 12:00hrs

Max Marks: 50

Instructions

- **Please start** the answer to each question on new page **ONLY** of your answer sheets
- Please write your **correct exam number** without fail **on the answer sheets** as well as **on the question paper**
- Please be brief and to-the-point in the attempting the answers to the theory questions. Unnecessary and unjustified elaboration will not fetch more marks. The hallmark of an answer to a short question indeed is its brevity

Q.1 Answer the following:

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|---|---|----|
| 1 | Explain following terms in detail(ANY FOUR): (a)Scope of control (b)Cloud Vs Container (c)SAML (d)CAP theorem with example (e) Web 2.0 vs Web 1.0 | 08 |
| 2 | Explain types of community cloud with their merits and demerits. | 04 |
| 3 | Define web services. Explain components of web services in brief. | 05 |
| 4 | Differentiate between attacks and threats. Explain any two attacks on virtualization layers in detail. | 04 |
| 5 | Draw and explain conceptual reference architecture of cloud computing in detail. | 04 |
| 6 | Differentiate between time shared and space shared scheduling with respect to Cloudsim. | 05 |

Q.2 Answer the following:

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|---|--|----|
| 1 | A guest OS running on Xen attempts to directly update its page table to remap virtual memory. The update fails, and the system logs show a protection fault. Based on Xen's paravirtualized architecture, why does the guest OS fail to update the page table directly, and how should it correctly perform the update? | 04 |
| 2 | A cloud provider is migrating a virtual machine using the pre-copy live migration technique. The VM hosts a latency-sensitive interactive application. During migration, users experience temporary performance degradation. Explain why this performance degradation may occur with pre-copy migration. How can this migration technique be optimized to reduce service disruption? | 04 |
| 3 | A genomics research lab is running long-running simulation experiments that occasionally crash due to unstable code. They also need to scale the simulation across multiple machines to process different input datasets in parallel. Which virtualization technique would you recommend for this use case and why? Justify your answer with reference to how the technique works and how it addresses the application's needs. | 04 |
| 4 | An organization wants to run two web applications on the same physical host without using virtual machines (VMs). Each application should be isolated in terms of filesystem, networking, and resource usage. Is this possible? If yes, explain how it can be achieved. | 04 |
| 5 | An organization is migrating its on-premises project management tool to AWS. The architecture must address the following:
(a) Compute & Scalability: <ul style="list-style-type: none">○ Recommend services for auto-scaling compute.○ Explain how they handle fluctuating traffic. (b) Storage & Database: <ul style="list-style-type: none">○ Suggest storage for static content and backups.○ Choose a managed database with high availability. (c) Networking & Security: <ul style="list-style-type: none">○ Design a secure VPC with subnets and security groups.○ Include encryption and access control. (d) Real-time Notifications: <ul style="list-style-type: none">○ Propose a service for sending real-time alerts.○ Describe its integration briefly. | 04 |